1 Overall Results

Table~1 compares the main finite-sample criteria with estimators shown across columns.

Table 1: Overall finite-sample performance

Measure	DML-style IVQR	Oracle IVQR	Mean-Lasso Post-selection IVQR
Coverage	94.90%	94.19%	92.42%
Bias	0.009	-0.007	-0.057
MAE	0.760	0.602	0.624
RMSE	0.987	0.836	0.859
Estimate SD	1.045	0.930	0.928
Mean CR length	2.959	2.407	2.380
Median CR length	3.188	2.235	2.199
Full-grid rate	33.09%	19.37%	18.22%
Unresolved rate	0.06%	0.00%	0.00%

1.1 Estimators

1. **Oracle IVQR** knows the true active support and isolates the cost of nuisance estimation and selection, but is infeasible outside simulation.
2. **Mean-Lasso Post-selection IVQR** selects controls from the data and then applies IVQR, providing a repository-specific feasible sparse comparison.
3. **DML-style IVQR** uses cross-fitted nuisance estimation and residualized scores to reduce first-order sensitivity to nuisance-learning error.

1.2 Overall-results discussion

The implemented DML-style IVQR procedure has the highest overall coverage (94.90%), but also the longest mean region (2.959). Oracle IVQR has the smallest MAE (0.602) and RMSE (0.836), making it the strongest point-estimation benchmark. Mean-Lasso Post-selection IVQR produces a slightly shorter mean region than Oracle (2.380 versus 2.407) but covers only 92.42%. The shorter set is therefore not evidence of superior inference.

The ranking changes with the criterion. Oracle performs best for point accuracy; the DML-style procedure performs closest to nominal average coverage in these simulations; Mean-Lasso Post-selection IVQR is slightly more compact than Oracle but inferentially less reliable.

2 Monte Carlo Design

Table~2 records the validated design grid and the implementation settings used by the completed full run.

Table 2: Validated Monte Carlo design

Component	Validated/configured value
DGPs	DGP1, DGP2, DGP3
Active support	DGP1: 5 active controls, DGP2: 10 active controls, DGP3: 5 active controls
Sample sizes	500, 1000
Control dimensions	200, 500
Instrument-strength index pi	0.1, 0.25, 0.5, 1.0
Quantiles tau	0.25, 0.5, 0.75
Replications per cell	500
Design cells	144
True alpha by tau=(0.25, 0.50, 0.75)	DGP1: (0.3255, 1.0000, 1.6745); DGP2: (0.3255, 1.0000, 1.6745); DGP3: (0.4371, 1.0000, 1.5629)
Initial alpha grid	[-1, 3], 21 points, spacing 0.2
CR refinement	Adaptive boundary refinement; tolerance 0.025
Nominal confidence level	95%
Critical-value multiplier	1
Estimators	DML-style residualized IVQR; true-support Oracle IVQR; Mean-Lasso Post-selection IVQR

The validated design contains 3 DGPs, 2 sample sizes, 2 control dimensions, 4 instrument-strength settings, and 3 quantiles. Their Cartesian product gives 144 design cells with 500 replications per cell, 72,000 replications per estimator, and 216,000 rows overall.

3 Coverage Across Design Cells

Aggregate coverage can conceal heterogeneous cell-level performance. Table~3 therefore summarizes the distribution of the validated coverage values across design cells before the subgroup comparisons.

Table 3: Distribution of resolved-replication coverage across design cells

Coverage-dispersion measure	DML-style IVQR	Oracle IVQR	Mean-Lasso IVQR	Post-selection
Number of design cells	144	144	144	
Minimum coverage	91.80%	91.00%	83.80%	
10th percentile	93.60%	92.40%	89.66%	
25th percentile	94.20%	93.00%	90.80%	
Median	94.80%	94.40%	92.80%	
75th percentile	95.79%	95.25%	94.00%	
90th percentile	96.20%	95.80%	95.00%	
Maximum	97.39%	97.40%	96.60%	
Share below 90%	0.00%	0.00%	11.11%	
Share below 92.5%	0.69%	11.81%	43.06%	
Share below 95%	52.08%	65.97%	88.19%	
Share at or above 95%	47.92%	34.03%	11.81%	

Overall averages conceal meaningful design-level heterogeneity. The implemented DML-style IVQR procedure has a lower-tail (10th-percentile) cell coverage of 93.60%, and 0.00% of its cells fall below 90%. Oracle IVQR shows mild finite-sample undercoverage in its lower tail, with a 10th percentile of 92.40%. Mean-Lasso Post-selection IVQR has both frequent and severe undercoverage: 11.11% of cells fall below 90%, and its minimum is 83.8%. Average calibration must therefore be assessed alongside lower-tail cell performance.

4 Results by Quantile

Figure 1 compares empirical coverage by structural quantile. The vertical bars show 95% Monte Carlo uncertainty around each pooled coverage estimate, while the dashed line marks the nominal 95% level.

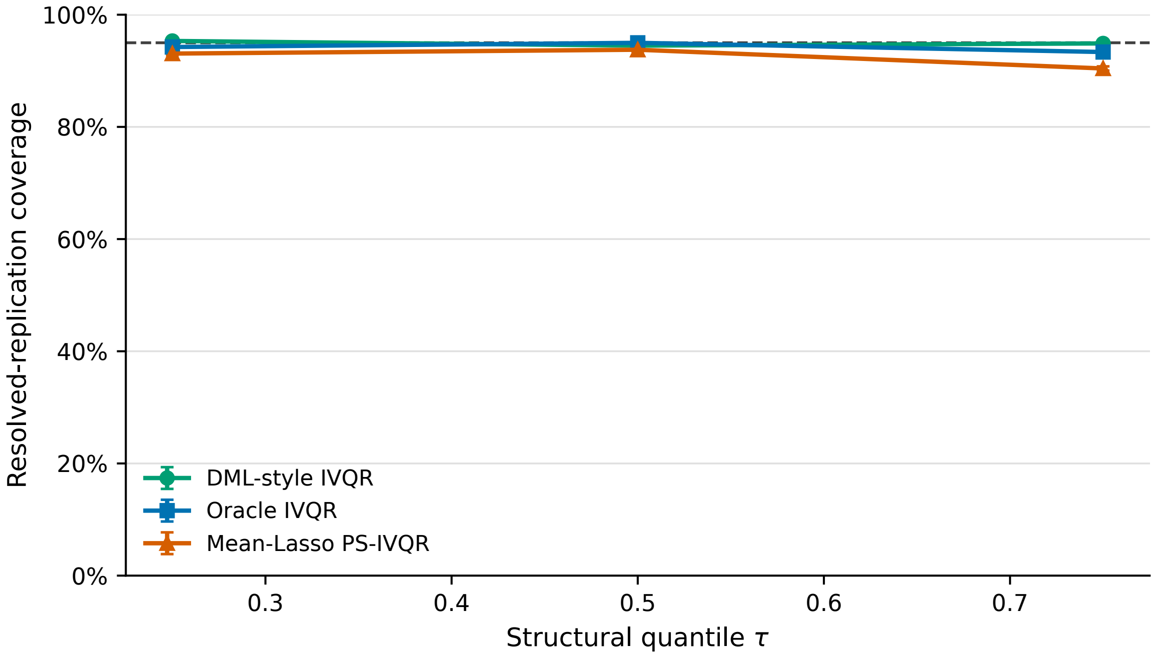

Figure 1: Empirical coverage by structural quantile. Error bars are normal 95% Monte Carlo intervals around pooled resolved-replication coverage; they do not establish uniform coverage across design cells.

Figure 2 and Figure 3 show the corresponding point- estimation behavior. Bias records direction, whereas RMSE combines dispersion and bias and should not be interpreted as an inferential-validity measure.

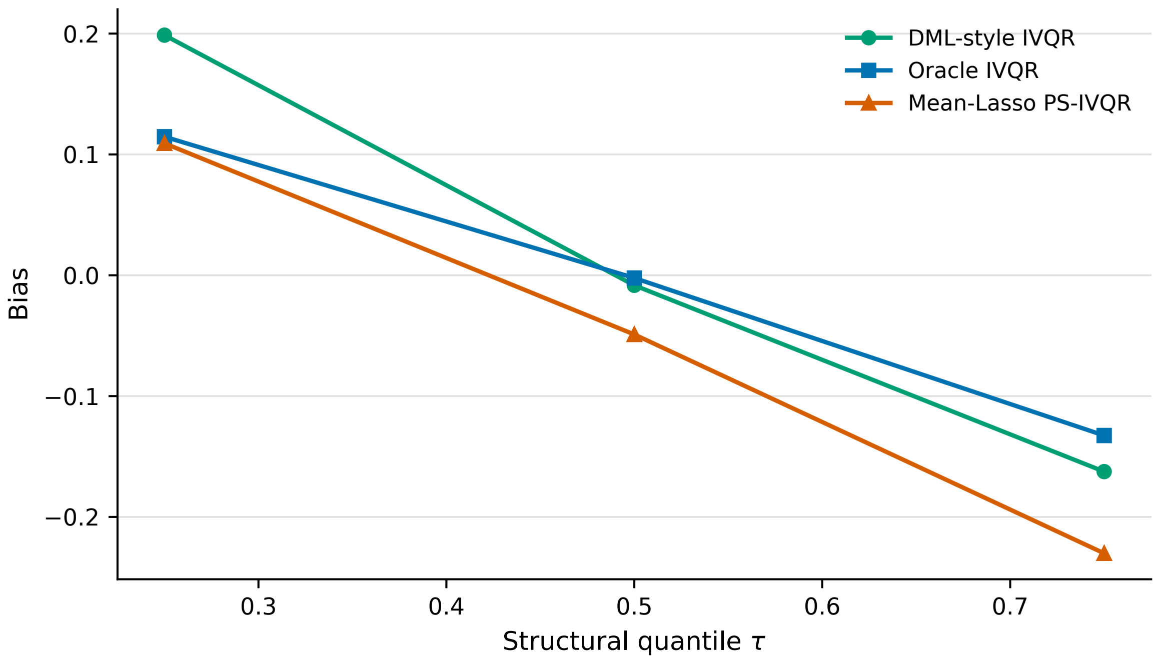

Figure 2: Bias by structural quantile. Values are computed from the validated quantile-level summary.

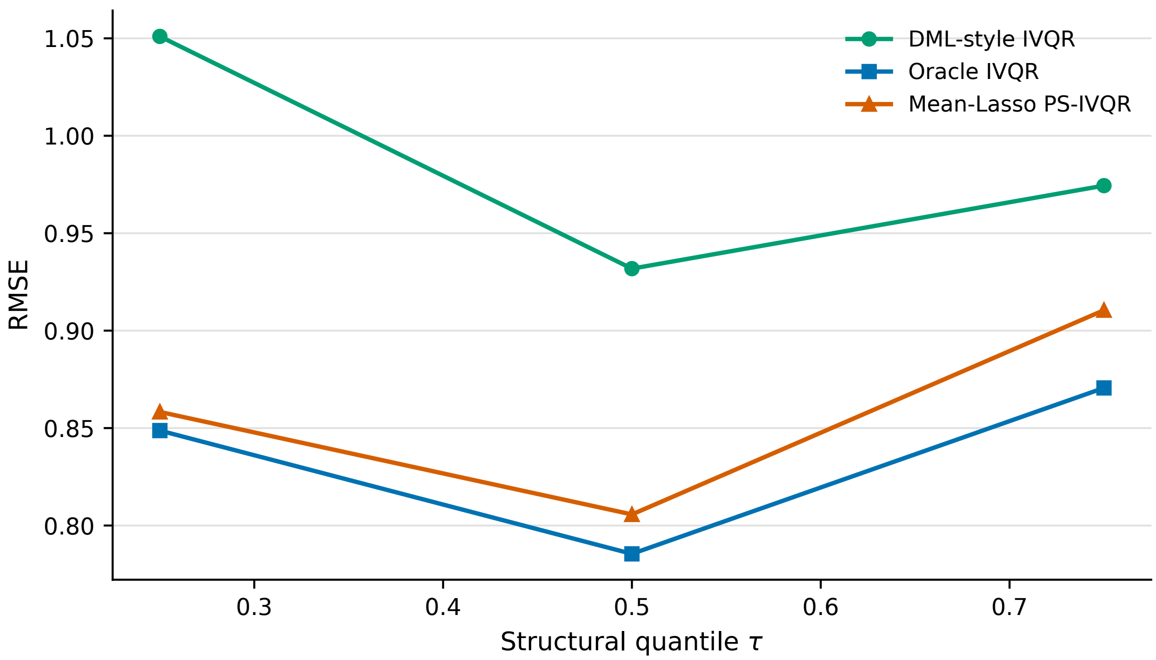

Figure 3: RMSE by structural quantile. Lower RMSE indicates stronger point-estimation performance but does not guarantee calibrated confidence regions.

Table~4 reports the underlying quantile-specific metrics with estimators arranged horizontally.

Table 4: Performance by structural quantile

Quantile and measure	DML-style IVQR	Oracle IVQR	Mean-Lasso IVQR	Post-selection
$\tau = 0.25$: Coverage	95.32%	94.25%	93.07%	
$\tau = 0.25$: Bias	0.199	0.115	0.109	
$\tau = 0.25$: MAE	0.821	0.621	0.631	
$\tau = 0.25$: RMSE	1.051	0.849	0.858	
$\tau = 0.25$: Mean CR length	3.031	2.423	2.372	
$\tau = 0.25$: Full-grid rate	33.45%	18.63%	17.25%	
$\tau = 0.25$: Unresolved rate	0.07%	0.00%	0.00%	
$\tau = 0.50$: Coverage	94.49%	94.99%	93.76%	
$\tau = 0.50$: Bias	-0.008	-0.002	-0.049	
$\tau = 0.50$: MAE	0.717	0.562	0.580	
$\tau = 0.50$: RMSE	0.932	0.785	0.806	
$\tau = 0.50$: Mean CR length	2.844	2.367	2.346	
$\tau = 0.50$: Full-grid rate	28.52%	18.55%	17.67%	
$\tau = 0.50$: Unresolved rate	0.05%	0.00%	0.00%	
$\tau = 0.75$: Coverage	94.88%	93.35%	90.42%	
$\tau = 0.75$: Bias	-0.162	-0.133	-0.230	
$\tau = 0.75$: MAE	0.741	0.622	0.660	
$\tau = 0.75$: RMSE	0.974	0.871	0.910	
$\tau = 0.75$: Mean CR length	3.002	2.429	2.421	
$\tau = 0.75$: Full-grid rate	37.30%	20.93%	19.73%	
$\tau = 0.75$: Unresolved rate	0.06%	0.00%	0.00%	

The main instability is concentrated in the upper quantile. Mean-Lasso Post-selection IVQR coverage is 90.42% at $\tau = 0.75$, with bias -0.230; its coverage is 93.07% at $\tau = 0.25$. Oracle also weakens at $\tau = 0.75$ but less severely, while DML remains close to nominal at each reported quantile. The coincident negative upper- quantile bias is consistent with, but does not by itself prove, a selection mechanism.

5 Results by Instrument Strength

Figure 4 shows coverage across the four instrument-strength settings with Monte Carlo uncertainty and the nominal reference line.

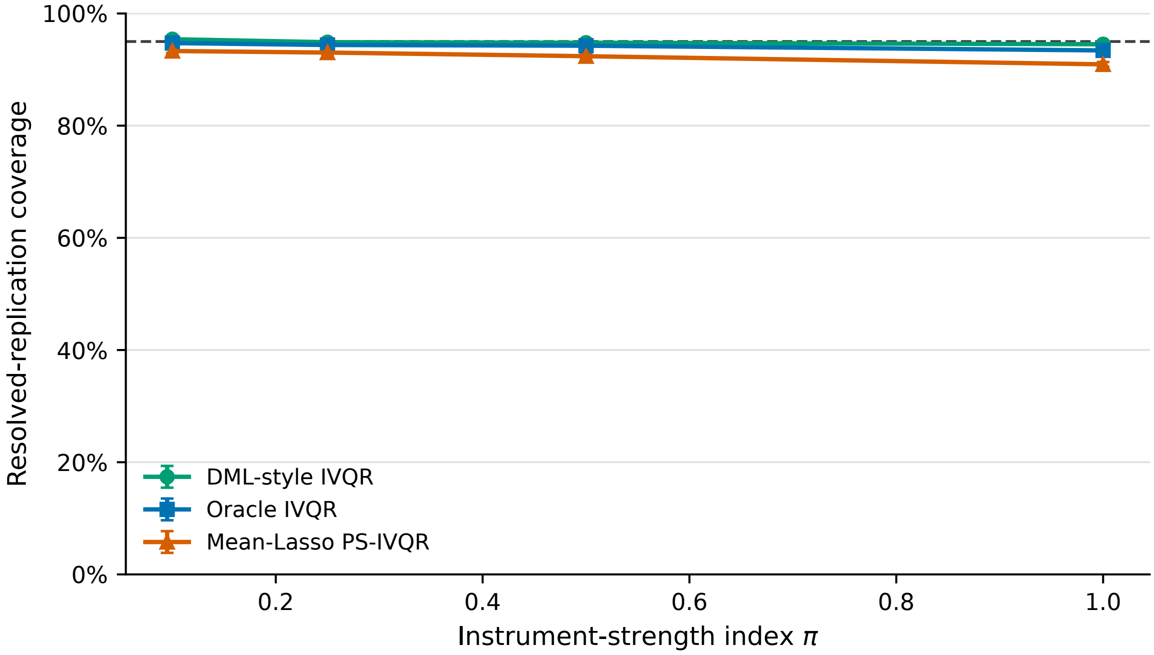

Figure 4: Empirical coverage by instrument-strength index. Error bars are normal 95% Monte Carlo intervals around pooled resolved-replication coverage; the coverage axis spans 0–100%.

Figure 5 and Figure 6 separate localization from calibration: shorter regions indicate more localization, while the full-grid rate shows how often the reported region accepts the entire parameter grid.

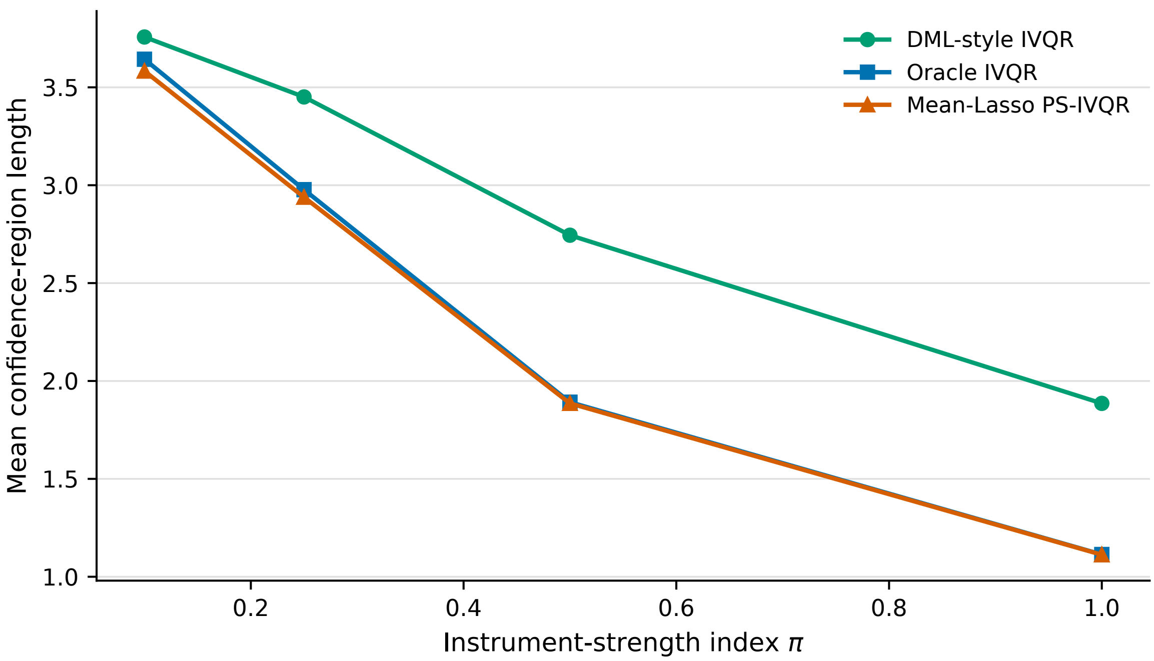

Figure 5: Mean confidence-region length by instrument-strength index. Region length is shown in parameter units, not percentages.

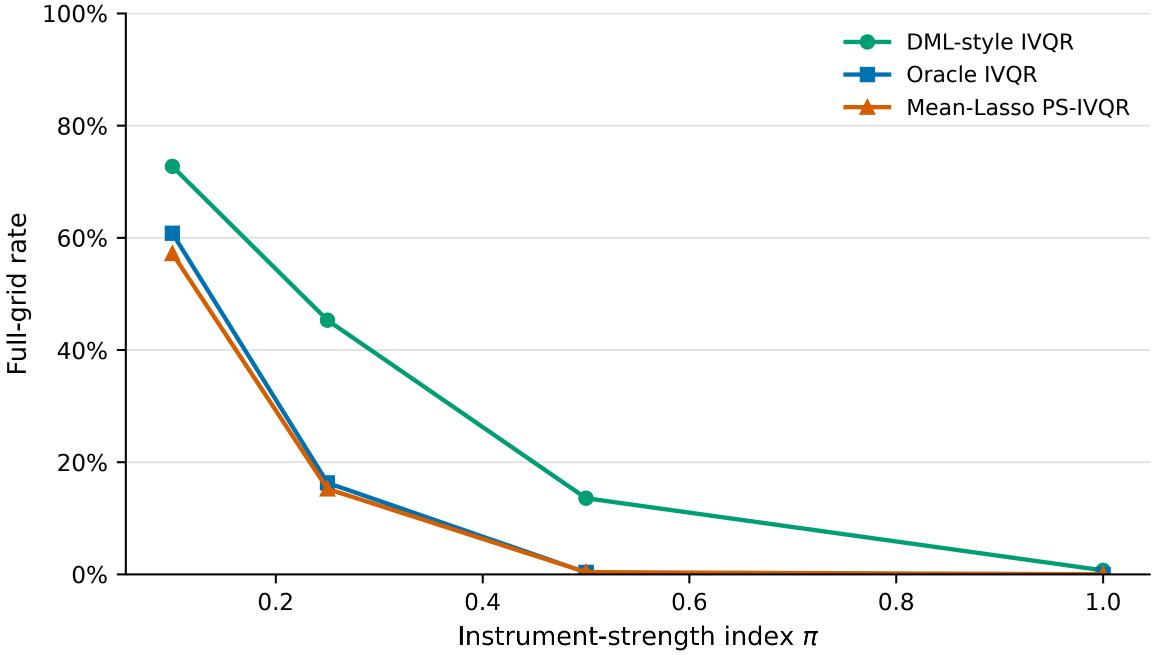

Figure 6: Percentage of replications with full-grid confidence regions by instrument-strength index.

Table~5 reports the validated strength-specific metrics with estimators as columns.

Table 5: Coverage and informativeness by instrument-strength index

Strength and measure	DML-style IVQR	Oracle IVQR	Mean-Lasso IVQR	Post-selection
$\pi = 0.10$: Coverage	95.41%	94.72%	93.31%	
$\pi = 0.10$: Bias	0.123	0.026	0.011	
$\pi = 0.10$: RMSE	1.315	1.281	1.290	
$\pi = 0.10$: Mean CR length	3.757	3.643	3.583	
$\pi = 0.10$: Full-grid rate	72.74%	60.82%	57.22%	
$\pi = 0.10$: Unresolved rate	0.13%	0.00%	0.01%	
$\pi = 0.25$: Coverage	94.88%	94.38%	93.03%	
$\pi = 0.25$: Bias	0.002	-0.035	-0.094	
$\pi = 0.25$: RMSE	1.121	0.891	0.932	
$\pi = 0.25$: Mean CR length	3.451	2.977	2.939	
$\pi = 0.25$: Full-grid rate	45.32%	16.32%	15.24%	
$\pi = 0.25$: Unresolved rate	0.08%	0.00%	0.00%	
$\pi = 0.50$: Coverage	94.79%	94.26%	92.38%	
$\pi = 0.50$: Bias	-0.047	-0.016	-0.086	
$\pi = 0.50$: RMSE	0.803	0.514	0.555	
$\pi = 0.50$: Mean CR length	2.745	1.892	1.886	
$\pi = 0.50$: Full-grid rate	13.58%	0.36%	0.40%	
$\pi = 0.50$: Unresolved rate	0.03%	0.00%	0.00%	
$\pi = 1.00$: Coverage	94.52%	93.42%	90.93%	
$\pi = 1.00$: Bias	-0.041	-0.001	-0.058	
$\pi = 1.00$: RMSE	0.516	0.306	0.335	
$\pi = 1.00$: Mean CR length	1.886	1.114	1.112	
$\pi = 1.00$: Full-grid rate	0.72%	0.00%	0.00%	
$\pi = 1.00$: Unresolved rate	0.00%	0.00%	0.00%	

All estimators become more informative as the instrument-strength index increases. For $\pi = 0.1$, mean CR lengths are 3.757 (DML-style), 3.643 (Oracle), and 3.583 (Mean-Lasso post-selection). At

$\pi = 1$, the corresponding lengths fall to 1.886, 1.114, and 1.112. This is the expected informativeness response to stronger identification. Mean-Lasso post-selection undercoverage, however, persists—and is most visible at $\pi = 1$—so weak identification is not its only plausible source.

6 Results by Sample Size and Dimension

Table~6 compares the four sample-size and control-dimension combinations.

Table 6: Performance by sample size and control dimension

Design and measure	DML-style IVQR	Oracle IVQR	Mean-Lasso IVQR	Post-selection
n=500, p=200: Coverage	94.72%	94.17%	92.46%	
n=500, p=200: Bias	0.022	-0.004	-0.049	
n=500, p=200: RMSE	1.034	0.897	0.921	
n=500, p=200: Mean CR length	3.123	2.606	2.577	
n=500, p=200: Full-grid rate	36.95%	23.08%	21.86%	
n=500, p=500: Coverage	94.94%	94.21%	91.77%	
n=500, p=500: Bias	0.022	-0.009	-0.092	
n=500, p=500: RMSE	1.073	0.892	0.933	
n=500, p=500: Mean CR length	3.238	2.597	2.572	
n=500, p=500: Full-grid rate	39.96%	22.78%	21.36%	
n=1000, p=200: Coverage	95.00%	94.14%	92.86%	
n=1000, p=200: Bias	-0.021	-0.005	-0.036	
n=1000, p=200: RMSE	0.824	0.777	0.783	
n=1000, p=200: Mean CR length	2.468	2.209	2.182	
n=1000, p=200: Full-grid rate	21.51%	15.59%	14.77%	
n=1000, p=500: Coverage	94.94%	94.26%	92.58%	
n=1000, p=500: Bias	0.015	-0.009	-0.049	
n=1000, p=500: RMSE	0.998	0.767	0.788	
n=1000, p=500: Mean CR length	3.008	2.215	2.189	
n=1000, p=500: Full-grid rate	33.94%	16.03%	14.88%	

Figure 7 adds Monte Carlo uncertainty to the same coverage comparison without introducing another performance metric.

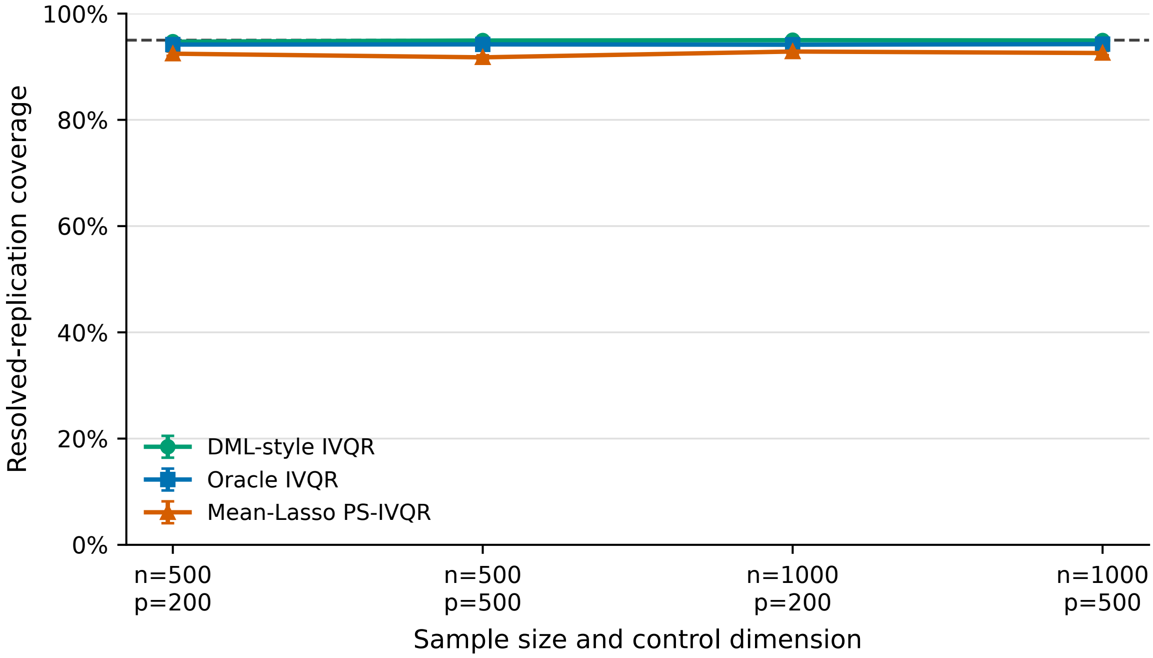

Figure 7: Empirical coverage by sample size and control dimension. Error bars are normal 95% Monte Carlo intervals around pooled resolved-replication coverage; the coverage axis spans 0–100%.

7 Coverage–Informativeness Trade-off

Figure 8 compares coverage and mean confidence-region length at each estimator–quantile aggregation. Within each estimator series, the connected points follow the quantiles from 0.25 through 0.75; the dashed line marks nominal coverage.

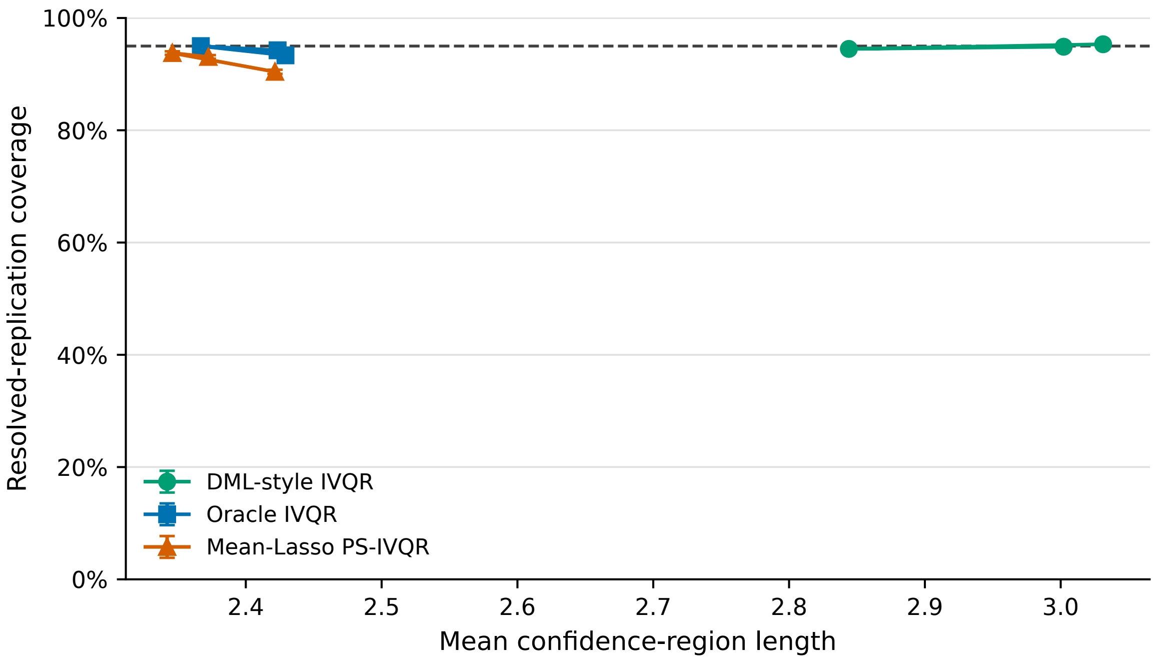

Figure 8: Coverage versus mean confidence-region length by estimator and structural quantile. Vertical bars show normal 95% Monte Carlo intervals. Shorter regions are preferable only when calibration remains credible.

8 Worst-Performing Design Cells

Tables~7 and~8 retain the ordinary row-oriented layout because every row is a distinct design cell. The two panels use identical row ordering to keep all requested fields readable.

Table 7: Ten lowest-coverage design cells: design and coverage

Estimator	DGP	n	p	pi	tau	Resolved	Coverage	MCSE
Mean-Lasso Post-selection IVQR	DGP2	500	500	1	0.75	500	83.80%	1.65%
Mean-Lasso Post-selection IVQR	DGP3	500	500	1	0.75	500	83.80%	1.65%
Mean-Lasso Post-selection IVQR	DGP2	500	200	1	0.75	500	85.80%	1.56%
Mean-Lasso Post-selection IVQR	DGP2	500	500	0.25	0.75	500	86.80%	1.51%
Mean-Lasso Post-selection IVQR	DGP2	500	500	0.5	0.75	500	87.00%	1.50%
Mean-Lasso Post-selection IVQR	DGP2	500	200	0.5	0.75	500	88.00%	1.45%
Mean-Lasso Post-selection IVQR	DGP2	1000	500	0.5	0.75	500	88.00%	1.45%
Mean-Lasso Post-selection IVQR	DGP2	1000	500	1	0.75	500	88.00%	1.45%
Mean-Lasso Post-selection IVQR	DGP1	1000	500	1	0.75	500	88.20%	1.44%
Mean-Lasso Post-selection IVQR	DGP2	1000	200	0.1	0.75	500	88.40%	1.43%

Table 8: Ten lowest-coverage design cells: performance and diagnostics

Estimator	DGP	n	p	pi	tau	Bias	RMSE	Mean CR length	Full grid	Unresolved
Mean-Lasso Post-selection IVQR	DGP2	500	500	1	0.75	-0.256	0.516	1.478	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP3	500	500	1	0.75	-0.157	0.391	1.171	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	500	200	1	0.75	-0.146	0.452	1.452	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	500	500	0.25	0.75	-0.472	1.279	3.244	35.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	500	500	0.5	0.75	-0.352	0.840	2.502	2.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	500	200	0.5	0.75	-0.270	0.825	2.513	3.60%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	1000	500	0.5	0.75	-0.154	0.570	1.836	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	1000	500	1	0.75	-0.107	0.339	1.008	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP1	1000	500	1	0.75	-0.086	0.290	0.940	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	1000	200	0.1	0.75	-0.402	1.327	3.560	59.00%	0.00%

The lowest-coverage cell is Mean-Lasso Post-selection IVQR, DGP2, $n = 500$, $p = 500$, $\pi = 1$, and $\tau = 0.75$, with coverage 83.8% and MCSE 1.65%. A 95% Monte Carlo interval is approximately [80.6%, 87.0%], far below 95%. Sampling noise from 500 replications cannot plausibly explain that gap.

Figure 9 shows how often the initial-grid point estimate falls on a boundary. It is a point-estimation diagnostic, not a coverage measure.

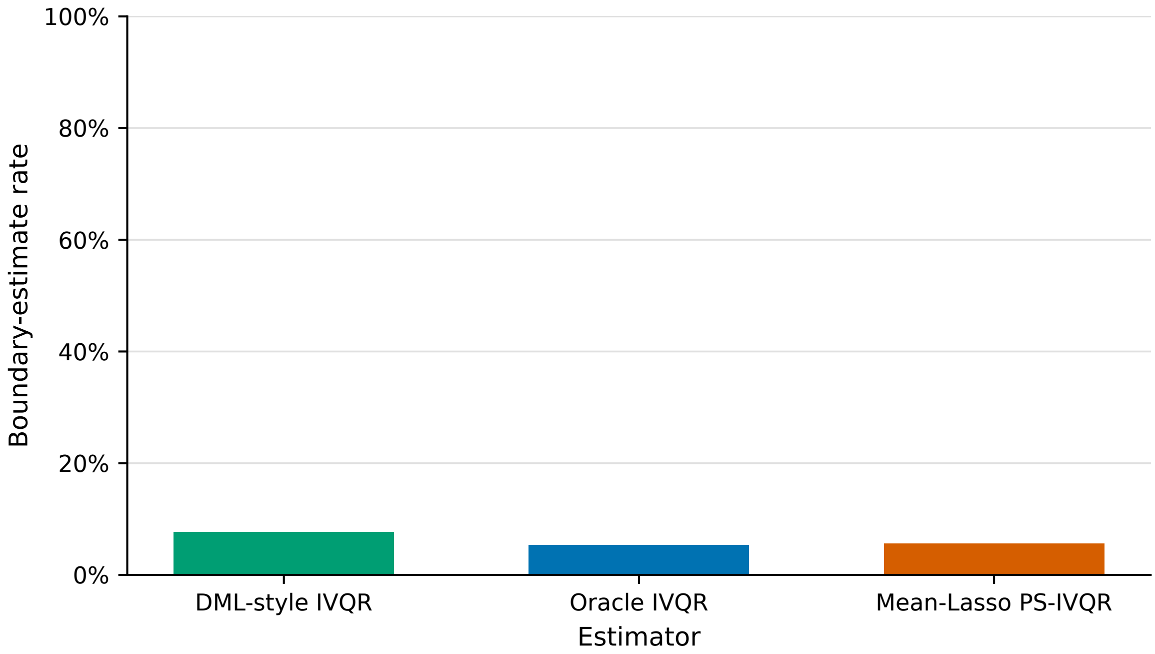

Figure 9: Boundary-estimate rate by estimator. Rates are based on the validated overall summary and are shown on a 0–100% axis.

The selected-control variable is available for Mean-Lasso Post-selection IVQR. Figure 10 presents it only as a descriptive diagnostic; variation in selection size is confounded with the underlying design dimensions.

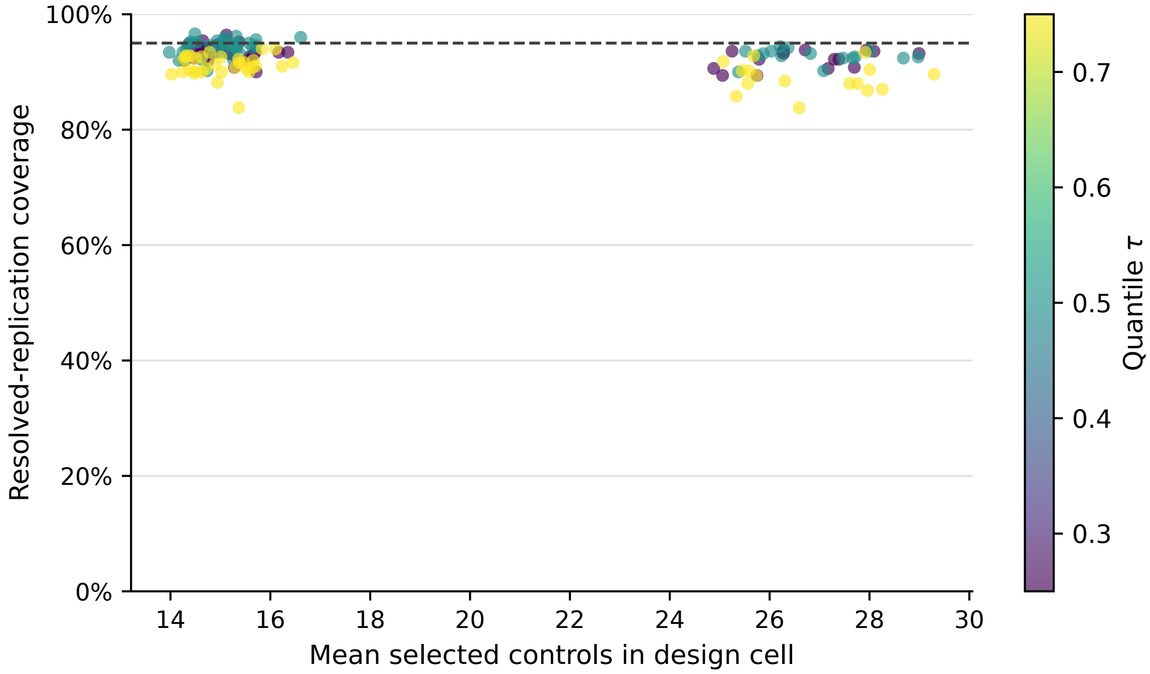

Figure 10: Mean-Lasso post-selection design-cell coverage versus the mean number of selected controls. Color denotes the structural quantile. The relationship is descriptive and does not identify a causal effect of selection size on coverage.